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In [3]: import os
os.environ["LOKY_MAX_CPU_COUNT"] = "4"  # silences the joblib warning

import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier, VotingClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay

FILE = r"D:\Datascience-MachineLearning\Datasets\diamonds.csv"
TARGET = "cut"
ORDER = ["Fair", "Good", "Very Good", "Premium", "Ideal"]

df = pd.read_csv(FILE)
df = df.loc[:, ~df.columns.str.contains("^Unnamed")]  # drop index column if present

X = pd.get_dummies(df.drop(columns=TARGET))  # color, clarity -> numbers
y = df[TARGET]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

tree = DecisionTreeClassifier(random_state=42)
forest = RandomForestClassifier(n_estimators=100, random_state=42)
ensemble = VotingClassifier(estimators=[("tree", tree), ("forest", forest)], voting="soft")

fig, axes = plt.subplots(1, 3, figsize=(18, 5))
for ax, (name, model) in zip(axes, [("Decision Tree", tree), ("Random Forest", forest), ("Ensemble", ensemble)]):
    model.fit(X_train, y_train)
    pred = model.predict(X_test)
    acc = accuracy_score(y_test, pred)
    cm = confusion_matrix(y_test, pred, labels=ORDER)

    print(f"{name} accuracy: {acc:.4f}")
    print(cm, "\n")

    ConfusionMatrixDisplay(cm, display_labels=ORDER).plot(ax=ax, cmap="Blues", values_format="d", colorbar=False)
    ax.set_title(f"{name}\nAccuracy: {acc:.3f}")
    ax.tick_params(axis="x", rotation=45)

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plt.tight_layout()
plt.savefig("confusion_matrices_diamonds.png", dpi=150, bbox_inches="tight")
plt.show()
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Decision Tree accuracy: 0.7178

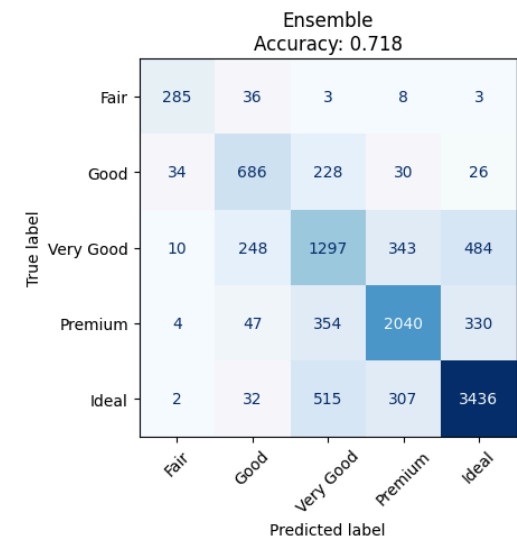
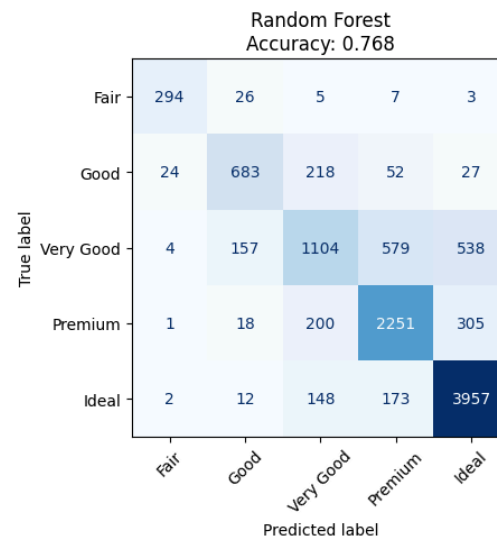
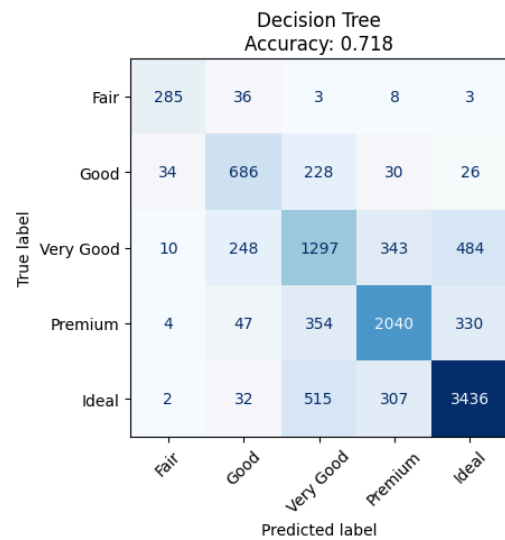
```
[[ 285   36    3    8    3]
 [   34  686  228   30   26]
 [   10  248 1297  343  484]
 [    4   47  354 2040  330]
 [    2   32  515  307 3436]]
```

Random Forest accuracy: 0.7684

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[[ 294   26    5    7    3]
 [   24  683  218   52   27]
 [    4  157 1104  579  538]
 [    1   18  200 2251  305]
 [    2   12  148  173 3957]]
```

Ensemble accuracy: 0.7178

```
[[ 285   36    3    8    3]
 [   34  686  228   30   26]
 [   10  248 1297  343  484]
 [    4   47  354 2040  330]
 [    2   32  515  307 3436]]
```



In []: